Project

Papers
1 https://hal.science/hal-02555457/

2 https://dl.acm.org/doi/pdf/10.1145/1150019.1136502

3 https://people.ece.ubc.ca/~sasha/papers/OSHMA2007.pdf

Comparison Table

<table>
<tr><td colspan="2">Software Implementation and Improvement</td><td>CUDA/GPU implementation</td><td>Hardware acceleration</td></tr>
<tr><td>CBC</td><td>CTR/GCM</td><td rowspan="2">It is faster than classical AES But is constrained by delay introduced because of transfer to data from RAM to VRAM and vice versa. Also it does not have specialised instructions for XOR and other operations.</td><td>AES NI</td></tr>
<tr><td>It is the classic AES implementation which is not parallelizable since there is a serial dependency of block Nth on (N-1)th block</td><td>These are the two modern parallelizable implementation of AES.
It encrypts a pseudo random counter which is then used for encrypting the data</td><td>Modern cpu have specialised hardware and ISA for AES operations. Instructions can be pipelined and executed in a single instruction.</td></tr>
</table>

1. Software Implementations & Algorithmic Bottlenecks

Cipher Block Chaining (CBC) Mode

Classical symmetric encryption frequently employs Cipher Block Chaining (CBC). In CBC encryption, the processing of block i strictly depends on the output of the preceding block $i - 1$:

$$C_i = E_K(P_i \oplus C_{i-1})$$

This feedback loop introduces a severe hardware bottleneck **data dependency stalls**. On modern CPU microarchitectures, an AES round instruction exhibits an execution latency of approximately 4 clock cycles, requiring ~ 40 clock cycles to complete a 10-round AES-128 block. Because block $i + 1$ cannot begin round 0 until block i completes round 10, the execution units sit idle, preventing instruction-level parallelization (ILP) within a single data stream.

Counter (CTR) and Galois/Counter Mode (GCM)

To bypass sequential dependencies, modern cryptosystems utilize Counter (CTR) mode or authenticated Galois/Counter Mode (GCM). CTR mode encrypts a sequence of deterministic counter blocks $IV \parallel i$ independently of the plaintext:

$$S_i = E_K(IV \parallel i), C_i = P_i \oplus S_i$$

Because counter blocks $S_1, S_2 … S_n have no mutual data dependencies, the algorithm exposes complete fine-grained parallelism. Modern CPUs can load multiple counter blocks into vector registers (e.g., 256-bit AVX2 or 512-bit AVX-512) and execute multiple AES rounds concurrently, converting the cryptographic latency bottleneck into a throughput-oriented task bounded only by system memory bandwidth.

2. GPU-Accelerated Cryptography (CUDA Paradigm)

General-Purpose Computing on Graphics Processing Units (GPGPU) leverages thousands of lightweight CUDA cores executing in a Single Instruction, Multiple Threads (SIMT) architecture. While GPUs provide massive theoretical compute capacity for parallel AES-CTR processing, they face two critical physical limitations:

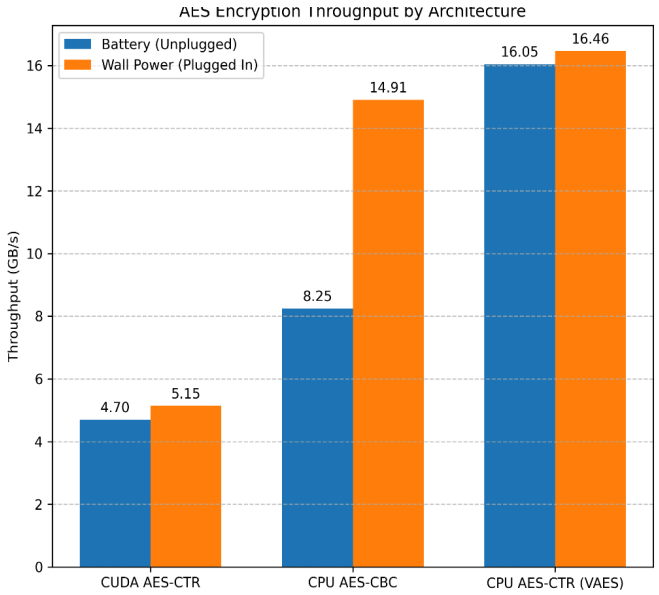
- **PCIe Interconnect Overhead:** Data must be transferred from host system RAM across the PCIe bus to discrete GPU VRAM and back. Even with asynchronous multi-stream memory copying (cudaMemcpyAsync), host-to-device bus latency often dominates total execution time, capping real-world throughput below host CPU rates for streaming workloads.
- **Lack of Native Cryptographic ISA:** Unlike modern CPUs, GPU Streaming Multiprocessors (SMs) generally lack specialized execution units for Galois Field $GF(2^8)$ math or native AES round instructions. GPU implementations must emulate AES by storing precomputed lookup tables (T-Tables) in on-chip Shared Memory or employing compute-heavy bit-slicing logic, incurring additional memory lookup and logic instruction overhead.

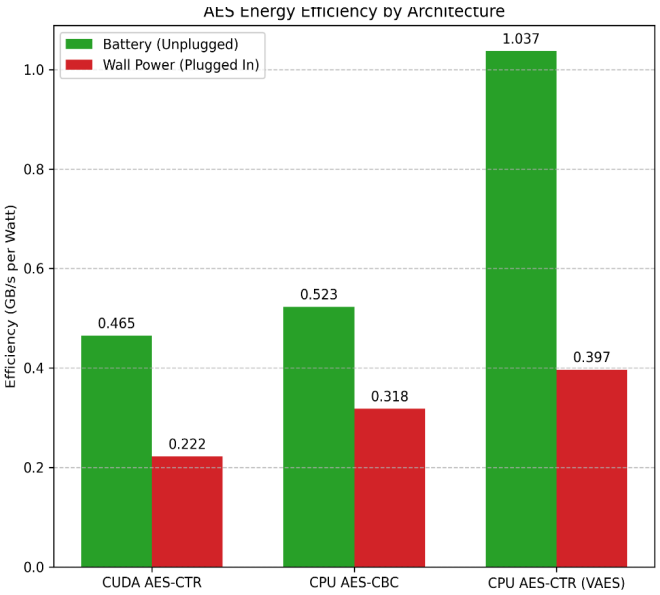
3. CPU Hardware Acceleration (AES-NI and Vector AES)

Modern x86 microarchitectures integrate dedicated hardware instruction set extensions—specifically **AES New Instructions (AES-NI)** and **256-bit/512-bit Vector AES (VAES)**:

- **Silicon Primitives:** Dedicated execution logic implements SubBytes, ShiftRows, MixColumns, and AddRoundKey directly in hardware circuitry via single instructions (e.g., AESENC, VAESENC).
- **Pipelined Execution:** Although an AESENC instruction has a multi-cycle latency, the execution units are fully pipelined with a throughput of 1 cycle. When combined with CTR mode, out-of-order execution engines interleave independent vector instructions to fully saturate hardware pipelines.
- **Zero Transfer Latency:** Because operations execute directly on CPU register files (XMM/YMM/ZMM) attached to host memory controllers, CPU-based hardware acceleration avoids the PCIe transfer bottlenecks inherent to discrete GPUs, offering superior throughput-per-watt efficiency for stream processing.

Implementation	Power State	Time	Energy	Power	Throughput	Efficiency
CUDA AES CTR	Battery	21.27	215.12	10.11	4.70	0.465
CUDA AES CTR	Plugged In	19.41	450.33	23.20	5.15	0.222
CBC	Battery	12.13	191.24	15.77	8.25	0.523
CBC	Plugged In	6.71	314.58	46.89	14.91	0.318
CPU AES CTR	Battery	6.23	96.47	15.49	16.05	1.037
CPU AES CTR	Plugged In	6.07	251.58	41.42	16.46	0.397





1. CPU AES-CTR (The Performance Benchmark)

- **Results:** Reached maximum system throughput (16.05–16.46 GB/s) and peak energy efficiency (1.037 GB/s per Watt on battery).
- **Architectural Reason:** Counter mode eliminates inter-block mathematical dependencies. This allows 256-bit Vector AES (VAES) instructions to process 8 blocks simultaneously per loop. Because data is modified directly inside CPU registers attached directly to the memory bus, execution hits the physical DDR system RAM bandwidth ceiling. On battery power, operating at lower clock frequencies minimizes dynamic power consumption ($P \propto V^2 f$), yielding maximum energy efficiency.

2. CPU AES-CBC (The Latency Bottleneck)

- **Results:** Throughput dropped by ~50% on battery (8.25 GB/s). Pushing throughput to 14.91 GB/s on wall power required an aggressive 46.89 W draw, dropping efficiency to 0.318 GB/s per Watt.
- **Architectural Reason:** CBC encryption forces block i to wait for block $i-1$ to complete all 10 AES rounds. This causes recurring pipeline stalls (~40 clock cycles per block) and disables AVX2/VAES vectorization within a stream. When plugged in, the OS frequency governor aggressive boosts CPU clock speeds to brute-force these latency stalls, causing energy consumption to spike disproportionately to performance gains.

3. CUDA AES-CTR (The Interconnect Bottleneck)

- **Results:** Delivered the lowest overall throughput (4.70–5.15 GB/s) and lowest energy efficiency (0.222–0.465 GB/s per Watt).
- **Architectural Reason:** The primary limitation is the physical PCIe bus. Transporting 100 GB of data from System RAM to VRAM and back ($H \to D \to H$) introduces massive latency that asynchronous streams cannot fully mask. Furthermore, discrete NVIDIA GPU Streaming Multiprocessors lack native dedicated AES hardware circuitry (requiring software T-Table emulation in shared memory) while adding the static baseline power consumption of the dedicated GPU board.

CUDA with CTR without power

Mode of Operation: CUDA AES-CTR (Shared Memory T-Tables)
Data Processed : 100 GB
Time Elapsed : 21.2683 seconds
Energy Consumed : 215.122 Joules
Average Power : 10.1147 Watts
Throughput : 4.70184 GB/s
Efficiency Metric: 0.464853 GB/s per Watt

CUDA with CTR with power

Mode of Operation: CUDA AES-CTR (Shared Memory T-Tables)
Data Processed : 100 GB
Time Elapsed : 19.4101 seconds
Energy Consumed : 450.334 Joules
Average Power : 23.201 Watts
Throughput : 5.15196 GB/s
Efficiency Metric: 0.222057 GB/s per Watt

Default scheduler with CBC without power

Mode of Operation: CBC (Cipher Block Chaining)
Data Processed : 100 GB
Time Elapsed : 12.1284 seconds
Energy Consumed : 191.236 Joules
Average Power : 15.7676 Watts
Throughput : 8.24512 GB/s
Efficiency Metric: 0.522914 GB/s per Watt

Default scheduler with CBC with power

Mode of Operation: CBC (Cipher Block Chaining)
Data Processed : 100 GB
Time Elapsed : 6.70868 seconds
Energy Consumed : 314.576 Joules
Average Power : 46.8909 Watts
Throughput : 14.9061 GB/s
Efficiency Metric: 0.317888 GB/s per Watt

Default scheduler without power

Data Processed : 100 GB
Time Elapsed : 6.22957 seconds
Energy Consumed : 96.4727 Joules
Average Power : 15.4863 Watts
Throughput : 16.0525 GB/s
Efficiency Metric: 1.03656 GB/s per Watt

Default scheduler with power

Data Processed : 100 GB
Time Elapsed : 6.0737 seconds
Energy Consumed : 251.583 Joules
Average Power : 41.4217 Watts
Throughput : 16.4644 GB/s
Efficiency Metric: 0.397483 GB/s per Watt

Custom AES algorithm on hybrid architecture

Workload Distribution (α)

For a buffer S_total, the data assigned to Performance (D_P) and Efficient (D_E) core pools across N_P P-threads and N_E E-threads is:

$D_P = \alpha \cdot S_{total}$
$D_E = (1 - \alpha) \cdot S_{total}$

The data assigned to each individual thread is:

$d_{p,i} = (\alpha \cdot S_{total}) / N_P$
$d_{e,j} = ((1 - \alpha) \cdot S_{total}) / N_E$

Cache Chunking (β_P, β_E)

The localized chunk size for step k in the execution loop for thread memory slices $d_{p,i}$ and $d_{e,j}$ is:

$B_{P,k} = \min(\beta_P, d_{p,i} - k \cdot \beta_P)$

$B_{E,k} = \min(\beta_E, d_{e,j} - k \cdot \beta_E)$

Only P cores AES CTR without power

Mode of Operation: AES-CTR (256-bit Vectorized)
Data Processed : 100 GB
Time Elapsed : 6.2599 seconds
Energy Consumed : 92.7753 Joules
Average Power : 14.8206 Watts
Throughput : 15.9747 GB/s

Efficiency Metric: 1.07787 GB/s per Watt

Only E cores AES CTR without power
Mode of Operation: AES-CTR (256-bit Vectorized)
Data Processed : 100 GB
Time Elapsed : 11.8062 seconds
Energy Consumed : 119.521 Joules
Average Power : 10.1236 Watts
Throughput : 8.4701 GB/s
Efficiency Metric: 0.836673 GB/s per Watt

Only P cores AES CTR with power
Mode of Operation: AES-CTR (256-bit Vectorized)
Data Processed : 100 GB
Time Elapsed : 6.03234 seconds
Energy Consumed : 199.037 Joules
Average Power : 32.995 Watts
Throughput : 16.5773 GB/s
Efficiency Metric: 0.502419 GB/s per Watt
Only E cores AES CTR with power
Mode of Operation: AES-CTR (256-bit Vectorized)
Data Processed : 100 GB
Time Elapsed : 6.59825 seconds
Energy Consumed : 164.419 Joules
Average Power : 24.9185 Watts
Throughput : 15.1555 GB/s
Efficiency Metric: 0.608204 GB/s per Watt

Theoretical Limit

Cache Optimization with P cores only without power
Mode of Operation: AES-CTR (In-Cache Vectorized)
Data Processed : 100 GB
Time Elapsed : 2.10003 seconds
Energy Consumed : 37.5451 Joules
Average Power : 17.8784 Watts
Throughput : 47.6185 GB/s
Efficiency Metric: 2.66347 GB/s per Watt

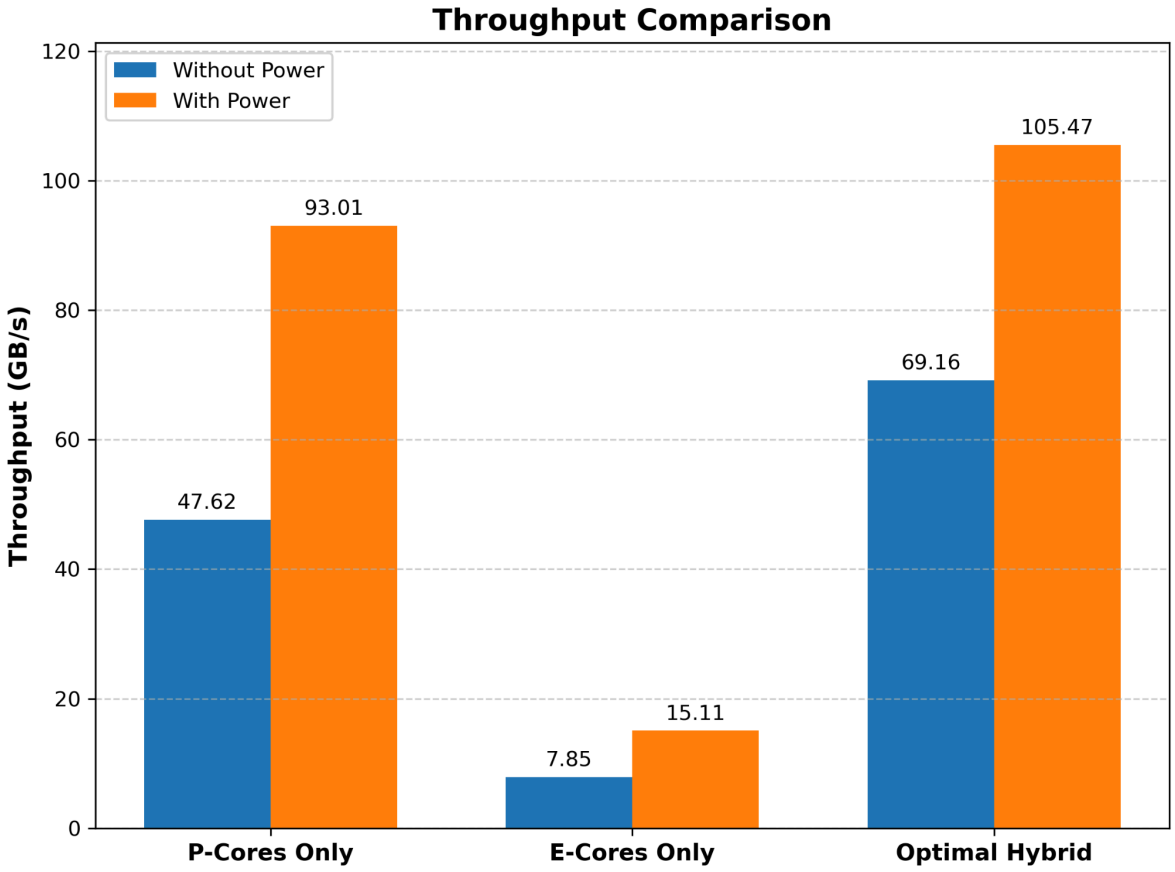
Cache Optimization with P cores only with power
Mode of Operation: AES-CTR (In-Cache Vectorized)
Data Processed : 100 GB
Time Elapsed : 1.07515 seconds
Energy Consumed : 59.5972 Joules
Average Power : 55.4316 Watts
Throughput : 93.0105 GB/s
Efficiency Metric: 1.67793 GB/s per Watt

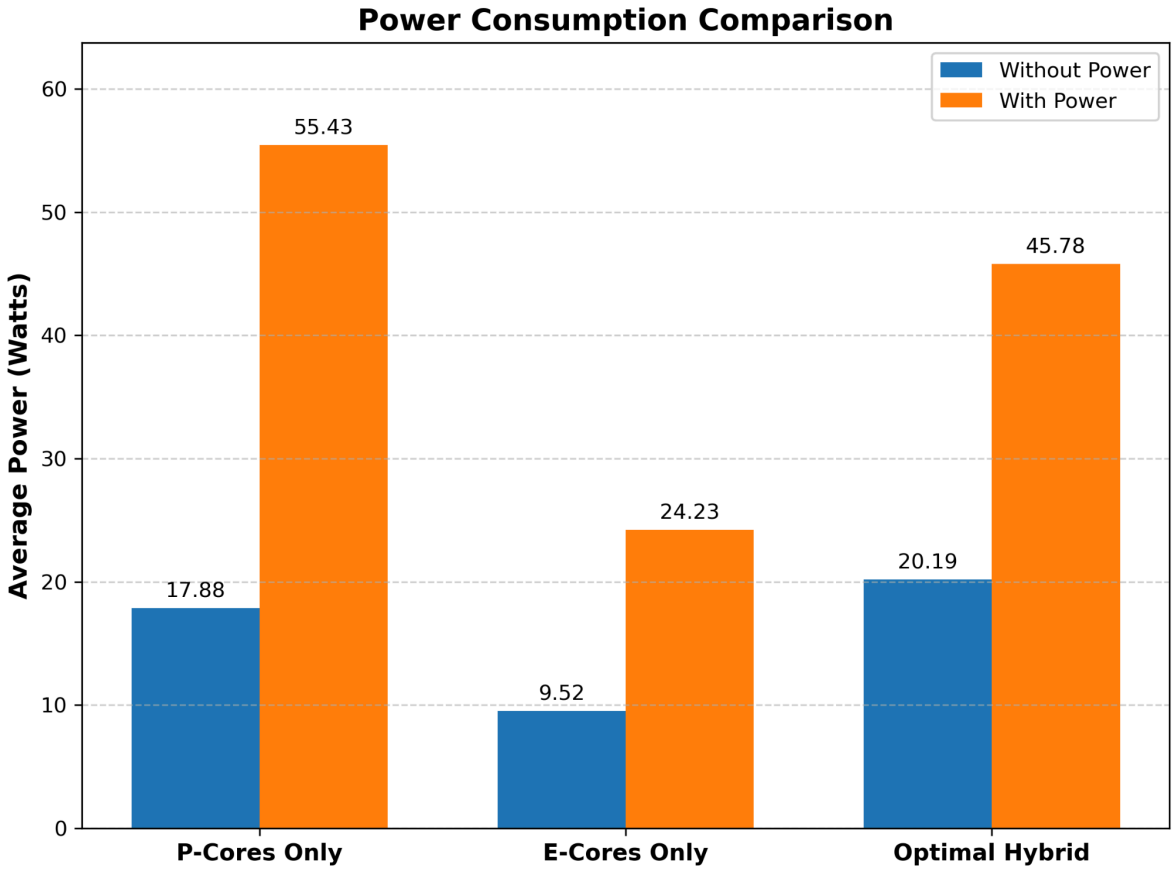
Cache Optimization with E cores only without power
Mode of Operation: AES-CTR (In-Cache Vectorized)
Data Processed : 100 GB
Time Elapsed : 12.7417 seconds
Energy Consumed : 121.26 Joules
Average Power : 9.51678 Watts
Throughput : 7.84825 GB/s
Efficiency Metric: 0.824675 GB/s per Watt

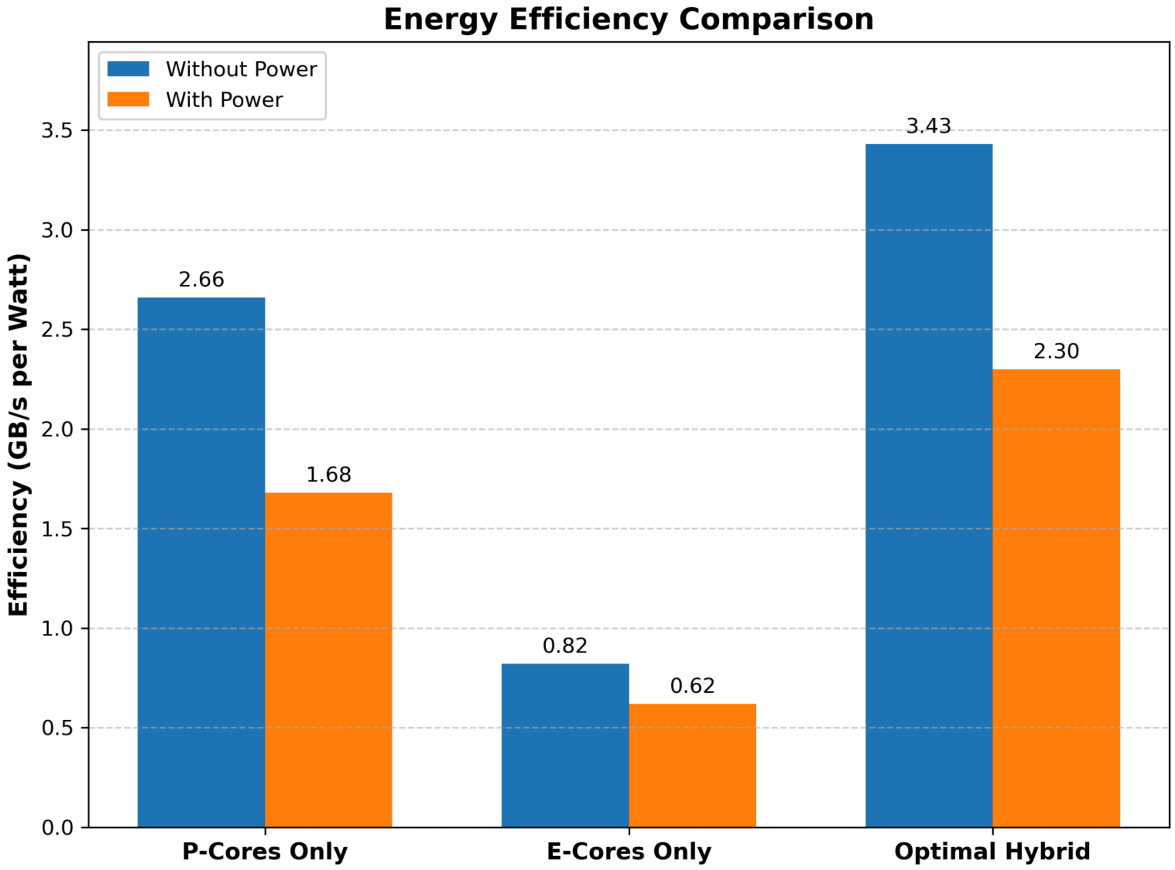
Cache Optimization with E cores only with power
Mode of Operation: AES-CTR (In-Cache Vectorized)
Data Processed : 100 GB
Time Elapsed : 6.61708 seconds
Energy Consumed : 160.355 Joules
Average Power : 24.2335 Watts
Throughput : 15.1124 GB/s
Efficiency Metric: 0.623617 GB/s per Watt

Cache Optimization with optimal parameters without power
Data Processed - 100GB
Time - 1.44583,
Energy Consumed - 29.1886,
Power - 20.1882,
Throughput - 69.1644,
Efficiency - 3.42599

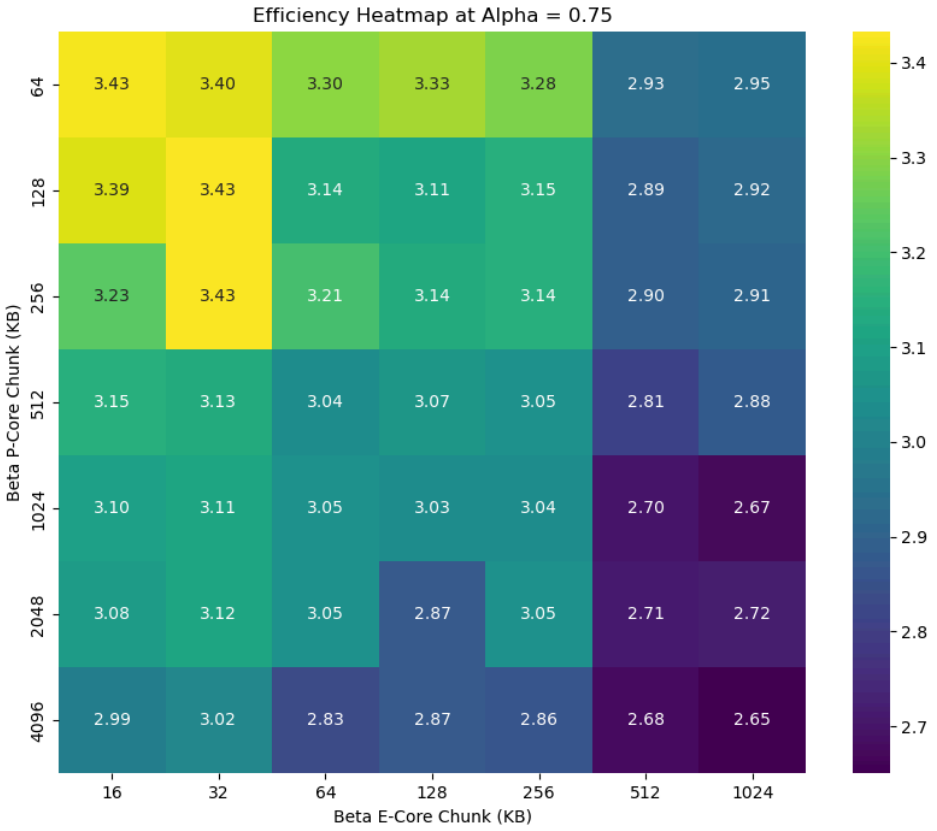
Cache Optimization with optimal parameters with power
Data Processed - 100GB
Time - 0.948116,
Energy Consumed - 43.4094,
Power - 45.7849,
Throughput - 105.472,
Efficiency - 2.30365

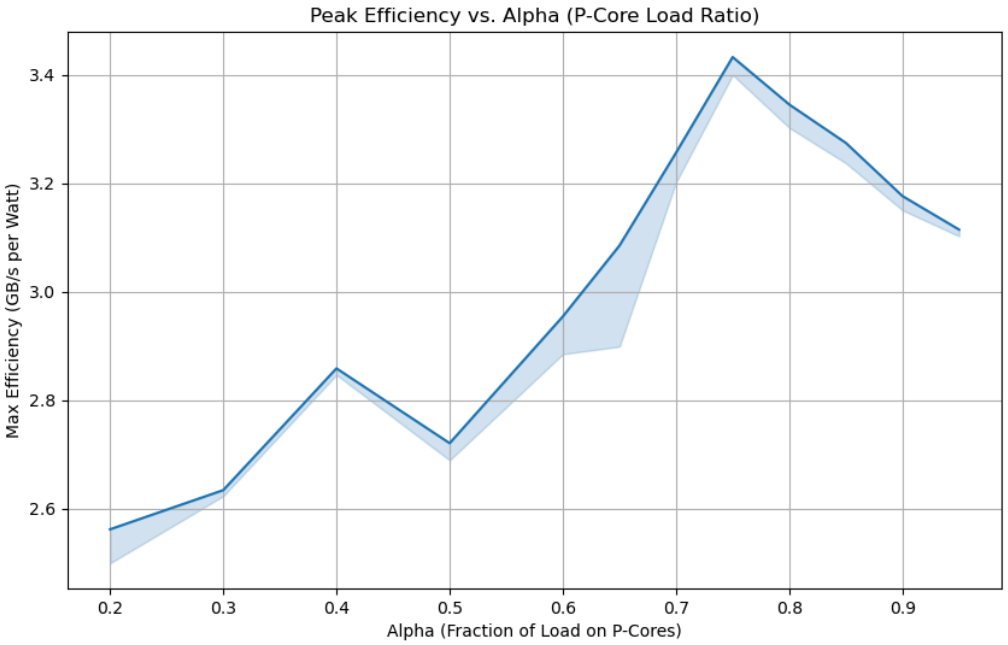

Throughput Comparison
Without Power
With Power
Throughput (GB/s)
120
100
80
60
40
20
0
47.62
93.01
7.85
15.11
69.16
105.47
P-Cores Only
E-Cores Only
Optimal Hybrid


Power Consumption Comparison
Without Power
With Power
Average Power (Watts)
0
10
20
30
40
50
60
17.88
55.43
9.52
24.23
20.19
45.78
P-Cores Only
E-Cores Only
Optimal Hybrid

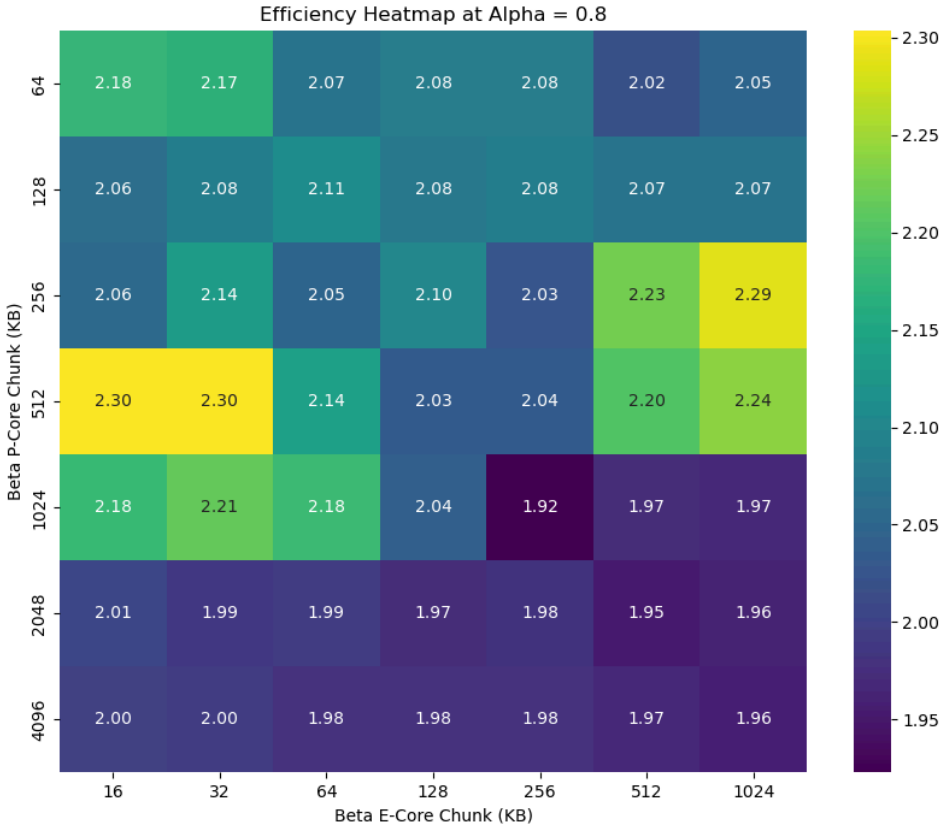

Energy Efficiency Comparison
Without Power
With Power
Efficiency (GB/s per Watt)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
2.66
1.68
0.82
0.62
3.43
2.30
P-Cores Only
E-Cores Only
Optimal Hybrid

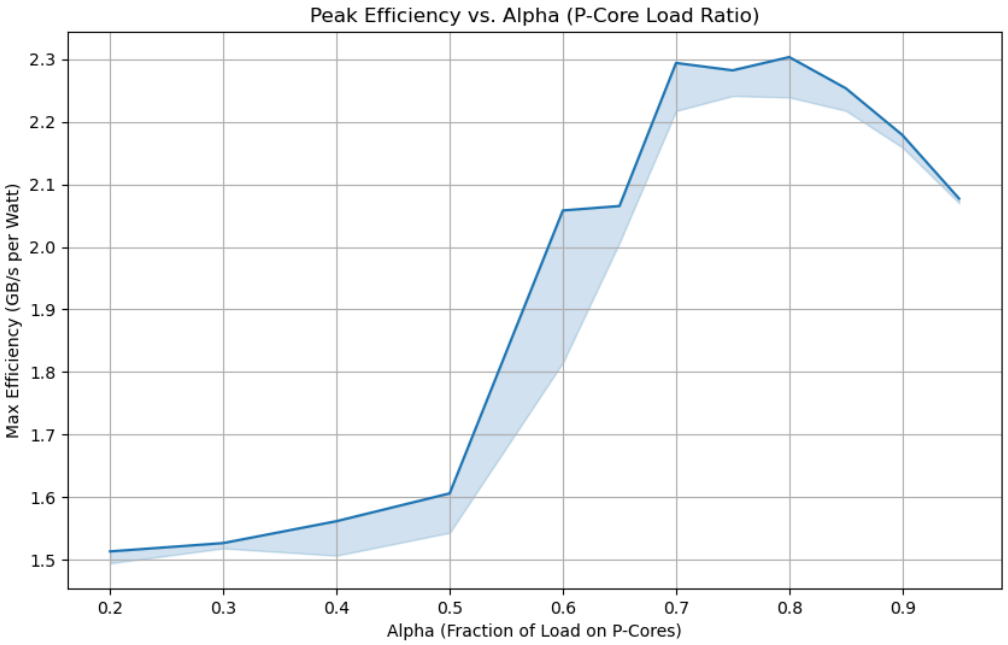
Graphs without Power (Theoretical limit with cache optimization)





Graphs with power (Theoretical limit with cache optimization)





Data Analytics

1. Alpha vs. Efficiency

- **Pearson Correlation (r):** 0.7985
- **p-value:** 2.93 \times 10^(-131)
- **Hypothesis Test:**
  - **1% Level:** Reject the null hypothesis ($p < 0.01$).
  - **5% Level:** Reject the null hypothesis ($p < 0.05$).
  - **10% Level:** Reject the null hypothesis ($p < 0.10$).
- **Conclusion:** There is a statistically significant, strong positive correlation between Alpha (P-core workload fraction) and Efficiency. As the share of data sent to the P-cores increases, efficiency predictably rises.

2. Beta_P vs. Efficiency

- **Pearson Correlation (r):** -0.0546
- **p-value:** 0.1862
- **Hypothesis Test:**
  - **1% Level:** Fail to reject the null hypothesis ($p > 0.01$).
  - **5% Level:** Fail to reject the null hypothesis ($p > 0.05$).
  - **10% Level:** Fail to reject the null hypothesis ($p > 0.10$).

- **Conclusion:** There is no statistically significant linear correlation between the P-core chunk size (beta_P) and Efficiency across the entire dataset.

3. Beta_E vs. Efficiency

- **Pearson Correlation (r):** -0.2985
- **p-value:** 1.43 * 10^(-13)
- **Hypothesis Test:**
    - **1% Level:** Reject the null hypothesis ($p < 0.01$).
    - **5% Level:** Reject the null hypothesis ($p < 0.05$).
    - **10% Level:** Reject the null hypothesis ($p < 0.10$).
- **Conclusion:** There is a statistically significant, moderate negative correlation between the E-core chunk size (beta_E) and Efficiency. Larger E-core chunk sizes generally lead to reduced efficiency.

4. Beta_P vs. Beta_E

- **Pearson Correlation (r):** approx 0.000
- **p-value:** 1.000
- **Hypothesis Test:**
    - **1% Level:** Fail to reject the null hypothesis.
    - **5% Level:** Fail to reject the null hypothesis.
    - **10% Level:** Fail to reject the null hypothesis.
- **Conclusion:** There is zero correlation between the P-core chunk sizes and E-core chunk sizes, confirming they were varied independently during testing (e.g., in a grid search parameter sweep).

Hardware-Accelerated Active Kernel Integrity and Cryptography on Hybrid

This project first develops a highly optimized, cache-aware scheduling algorithm (α, β_P, β_E) for hardware-accelerated cryptographic stream processing ($VAES$) on hybrid CPU architectures. These algorithmic findings are then deployed to construct a fast, power-efficient kernel lockdown and integrity verification system for Ubuntu

Technique	Efficiency	
	Without Power	With Power
CUDA	0.465	0.222
AES-CBC	0.523	0.318
E-cores only	0.825	0.624
P-cores only	2.663	1.678
Optimal	3.426	2.304

Kernel Lockdown and Integrity System

Module 1 (Extraction of files)

```
make[2]: Leaving directory
'/home/shikhar-shashank/Locked_Kernel/File_Extractor/files'
make[1]: Leaving directory '/usr/src/linux-headers-7.0.0-38-generic'
rmmod kernel_extractor 2>/dev/null
make: [Makefile:31: test] Error 1 (ignored)
insmod kernel_extractor.ko
./test_extractor; rc=$?; rmmod kernel_extractor; exit $rc
T1 header / geometry / alignment
  [PASS] magic + ABI version
  [PASS] ring geometry 1024 x 16384 B
  [PASS] descriptors 64B sized and aligned
  [PASS] payload slots 16 KiB aligned
  [PASS] frames tile 16-32 KiB (E) and 64-256 KiB (P) chunks
  [PASS] head/tail invariant
  [PASS] text bounds resolved
  [PASS] KASLR slide page aligned (slide=0x3b200000)
T2 mmap / open hardening
  [PASS] writable ring mapping refused
  [PASS] wrong-length mapping refused
  [PASS] MAP_PRIVATE refused
  [PASS] second opener gets EBUSY (SPSC preserved)
T3 one full scan epoch
  [PASS] epoch completed (2797 frames, 40.4 MiB, 376 module segments)
  [PASS] per-segment frame continuity (bad=0)
  [PASS] no unreadable pages (fault frames=0, fault_pages=0)
  [PASS] no writable PTEs inside kernel text (W^X)
  [PASS] .text bytes == _etext-_stext (24408304)
  [PASS] kernel saw no invalid tail values
T4 mapped addresses vs /proc/kallsyms
  [PASS] _stext matches
  [PASS] _etext matches
  [PASS] sys_call_table matches
T5/T6 sys_call_table integrity
  [PASS] 472/472 syscall entries inside kernel text (hooked entries: 0)
  [PASS] ring payload == live memory via /proc/kcore (3776 bytes)
```

Benchmarks

```
  producer      chunk=128 KiB :   9.15 GB/s (26169 MiB in 3.00s)
  ingest+AVX2 chunk= 32 KiB :   8.80 GB/s (25167 MiB in 3.00s) | frame age
p50 76.7 us p99 158.5 us max 1099.9 us (saturated ring) [x=33]
  ingest+AVX2 chunk=128 KiB :   9.34 GB/s (26713 MiB in 3.00s) | frame age
p50 67.2 us p99 128.4 us max 386.5 us (saturated ring) [x=6]
  ingest+AVX2 chunk=256 KiB :   9.44 GB/s (27018 MiB in 3.00s) | frame age
p50 64.0 us p99 126.9 us max 1508.8 us (saturated ring) [x=2f]
  kick -> first frame : p50 150.7 us  p99 248.7 us  max 329.1 us (n=100)

23 passed, 0 failed, 0 skipped | ring_full_events=888
shikhar-shashank:~/Locked_Kernel/File_Extractor/files$
```